

# Tool /framework assumptions — status delta

QDS assumptions\_report 36+ rows · 2026-05-05 re-run hardens R-free finding from 3-tool to 4-tool

| Assumption                                                | Was             | Now                                                             |
|-----------------------------------------------------------|-----------------|-----------------------------------------------------------------|
| ASSUM_interp_ion_identity_no_anomalous                    | known_violation | mitigated (CheckMyMetal)                                        |
| ASSUM_phenix_refine_in_run_r_factors_use_internal_scaling | known_violation | known_violation (4-tool hardened, NEW 05-05)                    |
| ASSUM_find_peaks_holes_cutoff_and_merging                 | known_violation |                                                                 |
| ASSUM_report_round_table_collapses_rounds                 | known_violation |                                                                 |
| ASSUM_report_position_initial_not_final                   | known_violation |                                                                 |
| ASSUM_report_water_count_stage_drift                      | known_violation |                                                                 |
| ASSUM_agg_zero_rama_rounds_one_outlier_to_zero            | mitigated       | mitigated (3-tool Asn A 39 hardens)                             |
| ASSUM_checkmymetal_geometry_based                         | —               | unchecked → mitigated by 1SAR run                               |
| ASSUM_t13_oracle_pending                                  | open follow-up  | closed — ctruncate wired                                        |
| ASSUM_aimless_requires_unmerged_intensities               | —               | mitigated — wrapper falls back to ctruncate                     |
| ASSUM_servalcat_sigmaa_per_shell_only (NEW 05-05)         | mitigated       | still mitigated · sigmaa direct R-work/R-free now also captured |

## Open assumption — K<sup>+</sup> alternative

K<sup>+</sup> alternative for both ions has the best raw bond-length fit but coordination is too low.

Will be added as an "unchecked" assumption if buffer content is investigated.

Trust model preserved: every load-bearing finding has ≥ 1 non-cctbx confirmation; 5 most consequential have ≥ 3.

Five findings with ≥ 3 confirmations: R-free · Asn A 39 · RMSD bracket · ligand RSCC · Mg<sup>2+</sup> exclusion.